Sample Project Documentation

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

Abstract

The Multimodal Attendance & Engagement Agent is an offline-first, edge-computing application that automates classroom attendance and evaluates student engagement using computer vision and audio signal processing. It detects and tracks faces, recognizes consenting students, estimates head pose and eye gaze, classifies classroom acoustics, fuses these signals into an engagement index, and gives the lecturer real-time teaching recommendations. All processing and storage remain on the local machine.

1. Introduction

Roll calls in classes of 60 to 120 students waste 5 to 10 minutes per lecture, proxy attendance is common, and lecturers lack objective feedback on attention. Cloud-based biometric systems add privacy risk. This project shows that a laptop and webcam can solve these problems locally.

2. Objectives

- Automate attendance using verified presence time.
- Measure engagement from gaze, pose, presence, and acoustics.
- Generate actionable teaching recommendations without alert fatigue.
- Protect privacy: offline processing, explicit consent, anonymous tracking.
- Provide a dashboard, database, and CSV export for lecturers.

3. Scope

In scope: single-camera classroom operation, 8-page dashboard, five-table SQLite database, and 12 automated test suites. **Out of scope:** measuring emotion or comprehension, multi-camera deployments, and cloud services.

4. System Design

A decoupled architecture is used. A FastAPI backend runs the vision, audio, fusion, and persistence layers and exposes REST, MJPEG video, and a 4 Hz WebSocket. A Streamlit dashboard presents eight pages: Overview, Live Monitoring, Attendance Management, Engagement Analytics, Teaching Recommendations, Student Session Reports, Data & Exports, and Settings.

Module	Technique	Output
Face detection	OpenCV Haar cascade (detectMultiScale3)	Boxes with calibrated confidence 0.50-1.00
Tracking	DeepSort + 128-d color/spatial embedding	Persistent track IDs
Recognition	DeepFace Facenet, cosine distance 0.15	Student ID or anonymous Student #N
Head pose / gaze	MediaPipe 478 landmarks + solvePnP; EAR below 0.16 means blink	Yaw, pitch, roll, gaze score
Audio	Librosa RMS, dBFS, ZCR, centroid, MFCC; adaptive noise floor	QUIET / SPEECH / MODERATE_NOISE / DISTURBANCE
Fusion	60 s TemporalSyncBuffer	Time-aligned observations

Module	Technique	Output
Engagement	0.40 gaze + 0.20 pose + 0.20 presence + 0.20 audio; EMA beta 0.70	Score and HIGH (0.70+) / MEDIUM (0.45-0.69) / LOW tiers
Attendance	Presence ratio = seconds present / session duration	PRESENT 75%+, PARTIAL 40-74%, ABSENT below 40%
Recommendations	5 rules, 30 s debounce	URGENT / WARNING / INFO cards with action steps

5. Implementation Summary

The project is written in Python 3.11 and organized into service modules (camera, detection and tracking, recognition, pose and gaze, audio, fusion, engagement, attendance, reasoning), a database package (SQLAlchemy models and CRUD), the FastAPI app, and the Streamlit dashboard. Fallbacks make it resilient: synthetic video without a camera, QUIET audio without microphone chunks, and bounding-box pose without the landmark model.

6. Testing and Results

All 12 test suites passed (100%). Engagement test profiles scored 0.968 (attentive), 0.598 (distracted), and 0.224 (low). Pose and gaze processing of 69 face poses gave a mean attentiveness of 0.72. The end-to-end test verified the full 11-step flow from health check to CSV export and database check.

7. Privacy and Ethics

- Zero cloud transmission: models and storage run locally.
- Enrollment raises a PermissionError without consent, so no embedding or photo is stored.
- No raw video is stored unless an administrator enables it.
- Deregistration purges the embedding, photo, and login.
- Manual overrides carry an audit reason. Engagement is a heuristic index, not a measure of understanding, emotion, or motivation.

8. Limitations

- Per-frame metrics are kept in memory and not saved to person_metrics.
- No built-in background microphone listener.
- Recognition threshold is conservative and sensitive to lighting.
- One camera cannot cover a very large hall.

9. Future Work

- Multi-camera RTSP support for auditoriums.
- Time-series storage (DuckDB or TimescaleDB) for per-frame telemetry.
- On-device small language model for natural-language lecture reviews.
- Student portal for viewing attendance and granting or revoking consent.

10. Conclusion

The project delivers a working, privacy-preserving attendance and engagement system that runs on commodity hardware. It meets its core objectives and passes all automated tests, with clearly identified next steps for real-world scale and benchmarking.